

UTM TENSILE REPORT — UTM_Test_20260905_152053

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

53.4 MPa

UTS

52.7 MPa

Yield σ_y

2.86 GPa

Modulus E

3.3 %

Failure ϵ_f

1551 kJ/m³

Toughness

306 N

Preload anchor

TEST SETTINGS

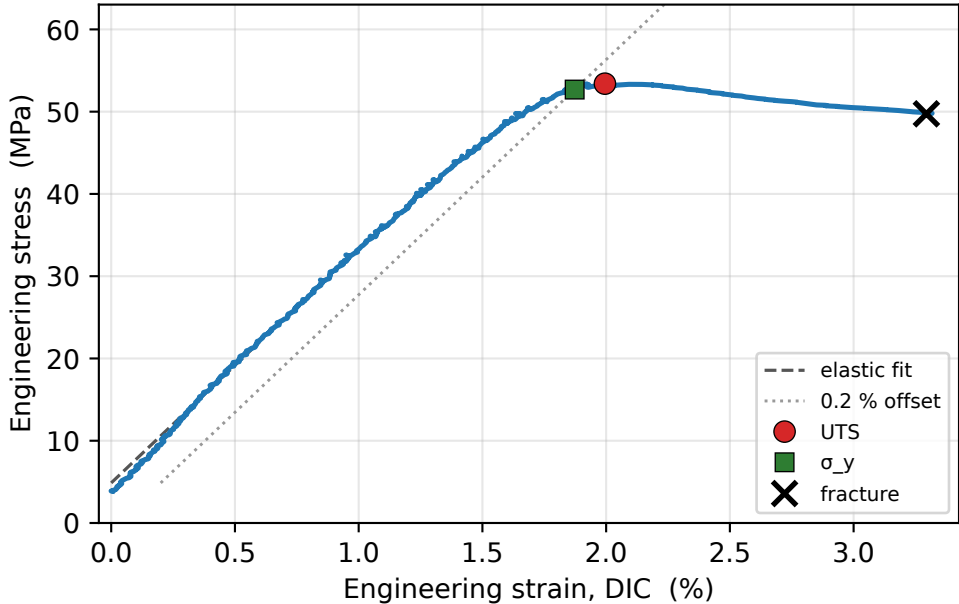
Specimen	UTM_Test_20260905_152053
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 330.947
Comment	—
Test date	2026-09-05 15:19:13
Duration	140.8 s

VALIDATION vs REFERENCES

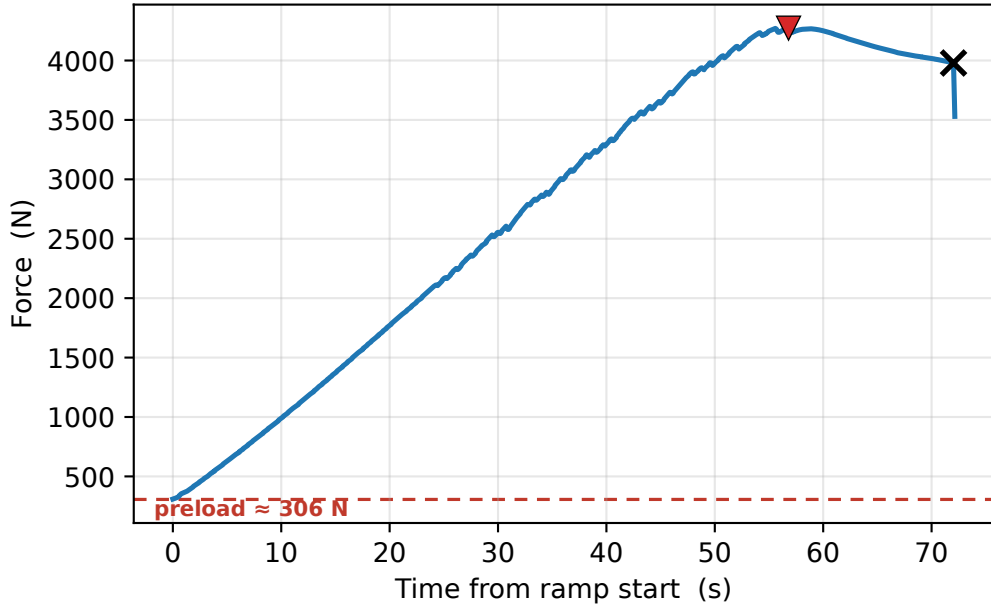
prop	meas	Chacón	E-PLA k
UTS	53.4	32-60 PASS	×1.09
σ_y	52.7	30-50 high	×1.10
E	2.9	3-5.5 low	×1.00
ϵ_f	3.3 %	(range 3-7 %)	×2.43

⚠ strength outside journal range

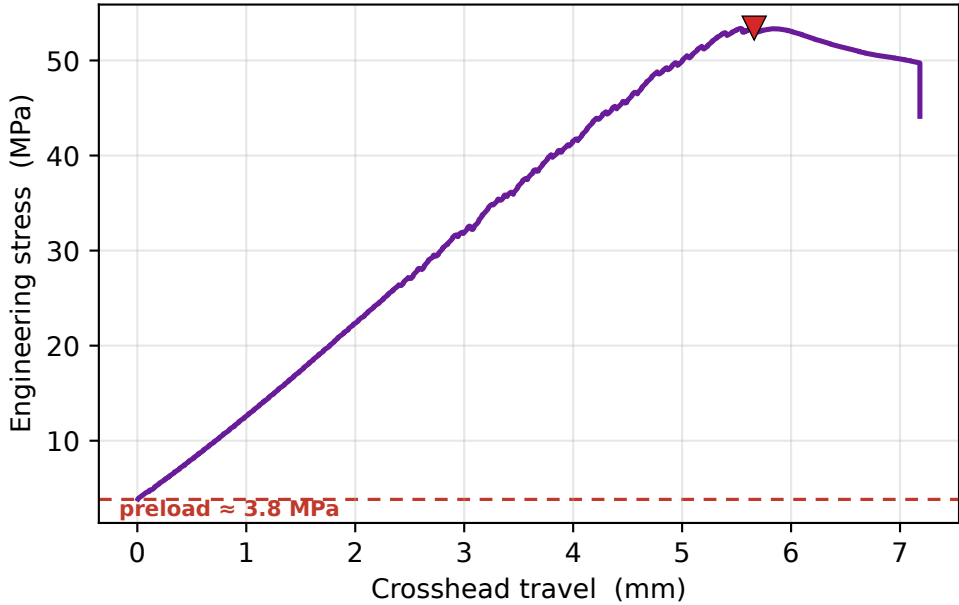
Stress-strain



Load vs time



Stress vs crosshead



DIC strain vs crosshead — the compliance gap

